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An Optimize Configuration of H-Bridge Multilevel Inverter

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***Abstract*— A traditional H-bridge inverter requires a minimum of four controllable switches and this multiplies with the levels in the inverter output voltage goes up. This work represents an improved multilevel inverter with a reduced number of switches. An improved structure using the sub-multilevel concept is proposed for a five-level inverter. This modified H-bridge inverter uses the same level of an inverter with a 25% reduction in the number of switches. This results in reduced switching losses, installation cost, and converter cost. Special care is taken to obtain an optimal number of switches. The proposed inverter is simulated for a five-level using Matlab/Simulink with phase disposition pulse width modulation (PDPWM) control for both resistive and resistive-inductive load. A comparison between conventional and modified H-bridge inverter is shown.**

Keywords—cascaded H-Bridge Multilevel Inverter, five-level inverter, modified 5-level inverter, THD

I. Introduction

A two-level inverter has two different output voltages and is the simplest one. But the ac output is rectangular with high total harmonic distortion (THD) whereas the load needs sinusoidal voltage. Multilevel inverter (MLI), a step ahead of the two-level inverter tends to reduce this effect. It does so by using many low rated dc voltage sources as input for the desired ac voltage output. So, in a multilevel inverter, the output voltage is stepped more than twice. As the level increases in the multilevel inverter, the waveform is smoother than the two-level inverter. MLI has two classes; current source inverter and voltage source inverter. In the former, a short circuit in the circuit can cause a very high fault current which will damage any other types of equipment connected to the circuit. Hence, multilevel voltage source inverters are preferred [2]. There are four types of MLI topologies; neutral point clamped (NPC), diode clamped (DC), flying capacitor (FC), and cascaded H-bridge (CHB) multilevel inverter. Cascaded H-bridge multilevel inverters have preferably low THD [1]. The difference between several topologies of MLI lies in the source of input voltage and the mechanism of switching.

MLI is used in power intense and high voltage applications. The industry demands additional solutions of high dv/dt resulting in voltage doubling in motor output, compliance of % THD, high electromagnetic interference (EMI) and high common-mode voltages which trigger research of inverters [3] [4]. Baker and Banister [5] pioneered the series H-bridge inverter. The demerits of DC and FC topologies, for instance, additional clamping diodes and capacitors are overcome by a proposed CHB inverter [6]. It can give high output power from a medium-voltage source. Higher voltage can be generated using the lower rating devices. The control of switches in this converter is simple and easy to construct. A survey on multilevel inverter configuration, control, and application, and an MLI with reduced number switches is described [6]. Because of its isolated dc sources, cascaded inverters are appropriate to interface photovoltaic generation to an ac grid for power quality management [7]. They are suitable for regenerative-type motor drive applications, fuel cell-based electric vehicles. Potential applications include electric and hybrid power trains. A review of MLI configuration was brought by Pharne [8]. An MLI with a reduction in the number of switches was described by Ebrahimi et al [9]. The topology has multiple bridge inverters with a sub-multilevel concept. Further development in MLI configuration's design and implementation was proposed by Najafi [10], Nedumgatt [11]. A CHB configuration with a reduction of switches count is described by Ebrahimi et al. [12]. A survey of multilevel inverter configurations, control, applications was made by Rodriguez et al. [13]. A CHB with regeneration capability and a smaller number of switches was dealt with by Lezana et al. [14]. In a similar context, Babaei [15] proposed CHB MLI for high-voltage applications.

Looking into the main advantages, owing to its modular arrangement and absence of energy-storing elements like capacitor a CHB MLI is used in this study. A new method is proposed as a modified H-bridge inverter having similar stepped output voltage using a smaller number of switches. This reduction in switches count will result in reducing switching losses, converter cost, and installation cost. Section 2 embodies the conventional and modified CHB structure for a 5-level inverter, followed by the modulation technique in section 3 and the simulation results, and finally, some conclusions are drawn.

II. 5-Level CHB Configuration

A. Conventional CHB MLI

Multilevel inverters are used in power conversion because of its output voltage and current have improved waveforms. For K sources in a CHB MLI then the output voltage will have m steps. The relations are

m=2K+1 (1)

No of source/No of bridges (K)=(m-1)/2 (2)

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No of switches(n)=(mX2)-2 (3)

Max voltage=KXV$_{dc}$ (4)

Here,

m=no of output voltage level

K=no of bridges

N=no of switching device

Fig.1, shows the CHB MLI with individual units in

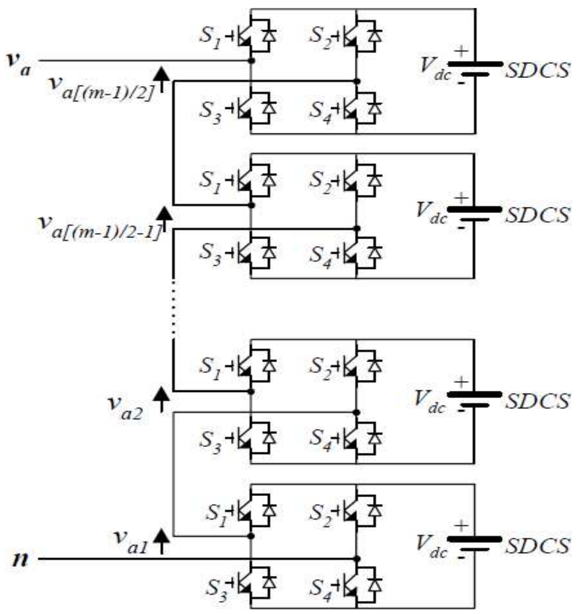

Fig. 1. Structure of cascaded H-bridge MLI

cascade that consist of separate dc sources (SDCS). From each dc source, ac output voltage is generated that has three steps +V_{dc}, 0, and –V_{dc}. Each H-bridge inverter consists of 4 switches normally. Conduction of S_1 and S_4 gives out voltage +V_{dc} whereas conduction of S_3 and S_2 gives out a voltage is –V_{dc}. When all the switches in the unit conduct, the output voltage is zero. As the output voltages are in series the MLI output aggregates the cell outputs.

B. Conventional 5-level CHB Configuration

The 5-level CHB inverter has two cascaded H-bridge units as illustrated in Fig. 2. As per (1)-(4), the cascaded 5-level bridge inverter uses 2 separate dc sources and 8 IGBTs. The output voltage is expressed as

$$V_o = V_{o1} + V_{o2} \tag{5}$$

Where V_{o1} and V_{o2} are the output voltages of the two units respectively. In a 5-level configuration the number of steps m is 5. The max voltage V_{max} is 2V_{dc} if both the units have identical sources of V_{dc}. The switching table for the switches in the H-bridge 5-level configuration in Table I.

TABLE I. SWITCHING STATE FOR THE 5-LEVEL CONVENTIONAL CONFIGURATION [16]

Modes	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8
2V_{dc}	on	off	off	on	on	off	off	on
V_{dc}	on	off	off	on	off	off	on	on
0	off	off	off	off	off	off	off	off
-V_{dc}	off	on	on	off	off	off	on	on
-2V_{dc}	off	on	on	off	off	on	on	off

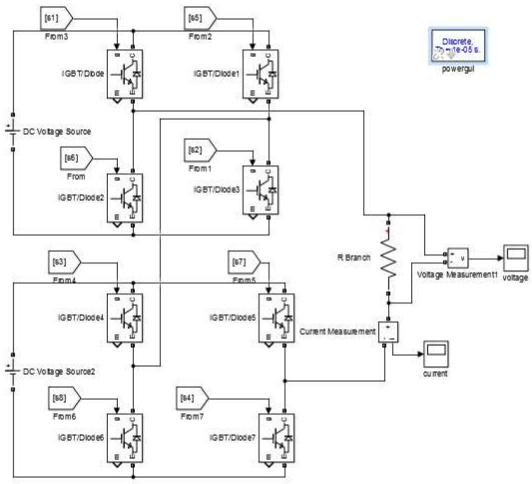

Fig. 2. Model of 5-level conventional CHB configuration

C. Modified Configuration

A new sub-multilevel inverter structure is illustrated herewith that uses a reduced count of IGBTs. Fig. 3 depicts its elementary configuration. It has a voltage source and two switches in place of four switches of a bridge. Both the switches cannot be on together at a time. If this happened

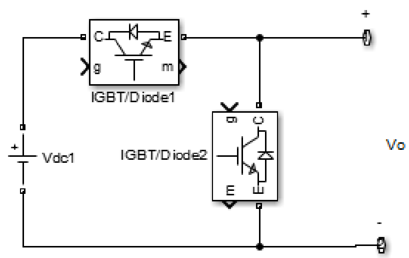

Fig. 3. Structure of sub-multilevel inverter

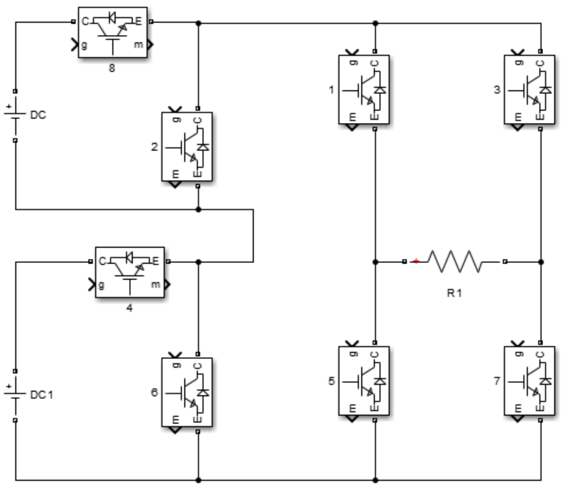

Fig. 4. Structure for generating full-cycle ac output voltages in sub-multilevel inverter

then a short circuit occurs. These sub-multilevel inverters joined in succession form a cascaded sub-multilevel inverter. The dc voltage sources associated with each sub-multilevel inverter have the same value. This structure has either zero and positive values of output. Fig. 4 shows the configuration which has both +ve and -ve values in output ac. This is because of the full-bridge topology added. S_1, S_2 are switched on together and conduct for $0<t<T/2$ and the output voltage is V_o. At $t=T/2$ S_1, S_2 are gated off and S_3, S_4 conduct for $T/2<t<T$ and the output voltage is $-V_o$, where T is the time period.

D. Modified 5-level CHB Configuration

The higher the voltage level, the no of switches required is higher. So, switching losses, costs, and complexity also increase. By using the modified multilevel inverter with less no. of switches the efficiency of the system increases. Fig. 5 shows the modified 5-level cascaded multilevel inverter which has 6 no. of IGBTs and 2 voltage sources. Here the output voltage step at $2V_{dc}$, V_{dc}, $-2V_{dc}$, $-V_{dc}$. Table II shows the on and off period of the switches of the modified 5-level inverter.

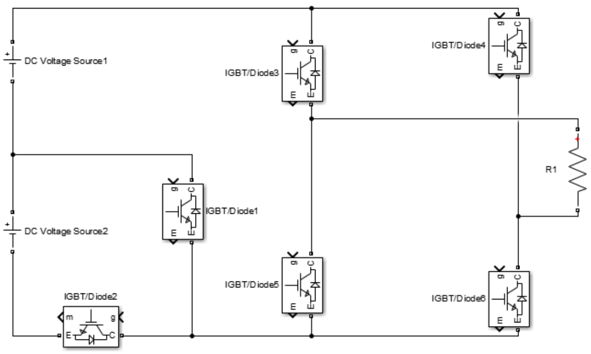

Fig. 5. Structure of modified 5-level inverter

TABLE II. SWITCHING STATE FOR THE MODIFIED 5-LEVEL CONFIGURATION

Modes	S_1	S_2	S_3	S_4	S_5	S_6
$2V_{dc}$	off	on	on	off	off	on
V_{dc}	on	off	on	off	off	on
0	off	off	off	off	off	off
$-V_{dc}$	on	off	off	on	on	off
$-2V_{dc}$	off	on	off	on	on	off

III. MODULATION TECHNIQUE

The switches of the multilevel inverter start working when they get a pulse. There are many techniques to generate the gate pulse. In this work, the pulse width modulation (PDPWM) technique is used. In this method, a sine wave is compared with the triangular wave and then produces the gate signal (Fig. 6). m-1 no. of the triangular wave is required (where m is the no of output voltage level). This method reduces unwanted harmonics from the output voltage.

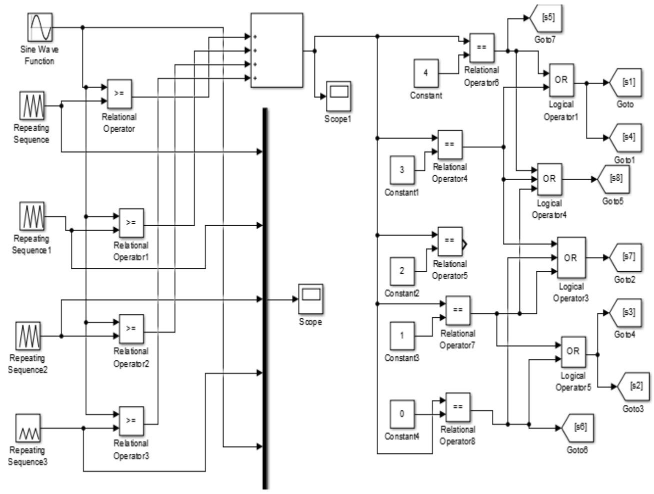

Fig. 6. Model of the PDPWM technique for 5-level cascaded inverter

This modulation imparts less THD for the line to line output voltages.

IV. SIMULATION RESULT

The conventional 5-level inverter and modified 5-level inverter are chosen for comparison of performance in this work. A 10 V dc power source is used along with a load resistance 10 Ω and an inductance 100µH. Both conventional and the modified topologies are simulated to judge their performance with resistive (R) and resistive-inductive (RL) load. With R load, the voltages are shown in

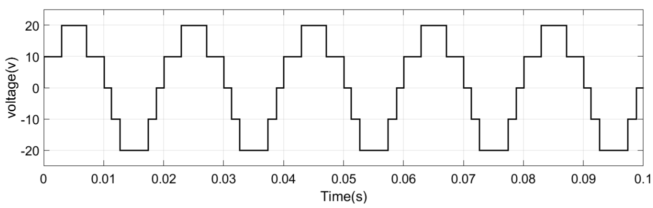

Fig. 7. Load voltage of conventional 5-level configuration

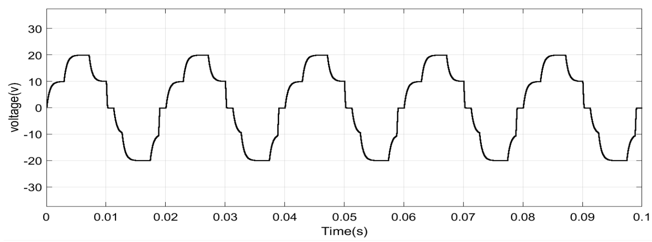

Fig. 8. Load voltage of modified 5-level configuration

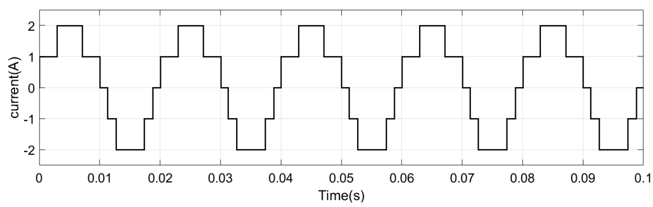

Fig. 9. Load current of conventional 5-level configuration

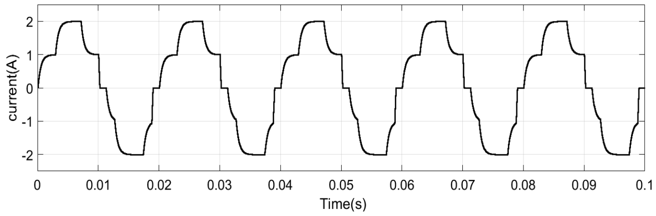

Fig. 10. Load current of modified 5-level configuration

Fig. 7 and Fig. 8 and the currents are shown in Fig. 9 and Fig. 10. Matlab/ Simulink was used to analyze output waveforms.

TABLE III. COMPARISON BETWEEN CONVENTIONAL AND OPTIMIZED5-LEVEL CHB MLI CONFIGURATIONS

Parameters	**Conventional**	**Modified**
No. of voltage source	2	2
No. of switches	8	6

TABLE IV. COMPARISON OF THD ANALYSIS BETWEEN CONVENTIONAL AND OPTIMIZED 5-LEVEL CHB MLI CONFIGURATIONS

THD (%)	**Conventional**		**Modified**	
	R load	***RL load***	***R load***	***RL load***
voltage	23.14	23.51	17.66	17.63
current	23.14	22.94	17.66	17.62

Simulation results presented in table IV for 5-level CHB MLI show that the modified configuration uses a lesser number of switches than the conventional H-bridge configuration and has similar performance for different types of load.

V. CONCLUSION

An improved structure using the sub-multilevel concept is proposed for a 5-level inverter. This modified H-bridge inverter uses the same level of an inverter with a 25% smaller count of switches. This subsides the cost, switching losses, and electromagnetic interference. However, the higher level of modified topology can significantly reduce the harmonics with a considerable saving of component cost and improvement of reliability in comparison to conventional CHB MLI. The optimized topology of the CHB MLI can be a good option in standalone single-phase PV power applications.

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